



A

MINOR PROJECT SYNOPSIS

Burnlytics

(A Smart Productivity & Burnout Detection System)

**Submitted in partial fulfilment of the requirements for the award
of the degree of B.Tech in Computer Science and Engineering**

Submitted By: Snigdha Mishra (23BCON1900)

Nikita Jain (23BCON1281)

Under the Supervision of

Ms. Priyanka Khatana

Designation of Guide

Department of Computer Science & Engineering

Session: 2025-26

JECRC University, Jaipur

TABLE OF CONTENTS

S. No.	Content	Page No.
1	INTRODUCTION	1
2	FEASIBILITY STUDY	2-3
3	PROJECT FEATURES	4-5
4	TOOLS, TECHNOLOGIES AND SYSTEM REQUIREMENTS	6-7
5	WORK PLAN & METHODOLOGY	8
6	CONCLUSION	9

1. INTRODUCTION

In today's fast-evolving digital and corporate environments, organisations heavily rely on efficient task management systems to ensure productivity and timely delivery of work. However, with the increasing complexity of workflows and the rising expectations from employees, individuals are often burdened with multiple tasks of varying priorities. This continuous pressure leads to workload imbalance, reduced efficiency, mental fatigue, and ultimately burnout, which negatively impacts both individual performance and organisational outcomes.

Despite the availability of modern task management platforms such as ServiceNow, most systems primarily focus on task assignment, tracking, and completion. They lack the capability to analyse workload patterns in real time or assess the well-being of employees based on their work behaviour. As a result, managers often fail to identify early signs of stress or overload, leading to decreased productivity and higher employee attrition rates.

To address these challenges, the Smart Personal Productivity & Burnout Detection System is proposed as an intelligent, workflow-driven solution. This system goes beyond traditional task tracking by incorporating a data-driven approach to monitor workload, evaluate productivity trends, and detect burnout risks proactively. It continuously collects and analyses key performance indicators such as the number of active tasks, task priority levels, average completion time, and backlog growth.

Based on these parameters, the system calculates a dynamic Burnout Score, which serves as a quantitative measure of an individual's workload pressure. This enables early identification of high-risk situations before they escalate into serious productivity or health issues.

Furthermore, the system leverages automation features such as real-time alerts, escalation workflows, and interactive dashboards to ensure timely intervention and better workload distribution.

The primary aim of this project is to enhance productivity while maintaining employee well-being by providing actionable insights and intelligent automation. By integrating monitoring, analysis, and response mechanisms into a single platform, the system contributes to creating a more balanced, efficient, and sustainable work environment.

2. FEASIBILITY STUDY

Need: In modern organisations, employees handle multiple tasks with varying priorities and tight deadlines, which often leads to workload imbalance, stress, and reduced productivity. While platforms like ServiceNow manage task assignment efficiently, they lack the ability to analyse workload pressure or detect early signs of burnout. As a result, managers cannot intervene in time, highlighting the need for a system that monitors workload intelligently and ensures a balanced, sustainable work environment.

Significance: The proposed system introduces a smart and proactive approach to productivity management by combining monitoring, analysis, and automation. It enhances transparency in workload distribution and provides data-driven insights into employee performance.

Key contributions include:

- Early detection of burnout risks using a dynamic scoring mechanism
- Improved workload balancing across teams
- Increased productivity through better task management
- Reduction in employee stress and fatigue
- Availability of real-time dashboards and alerts for quick decision-making

By integrating these features, the system ensures that productivity is achieved without compromising employee well-being, making it highly relevant in today's work culture.

Feasibility:

- **Technical Feasibility:** The system can be effectively developed using the ServiceNow platform, which provides built-in capabilities such as:

Database management (tables for tasks, users, and metrics)

Workflow automation using Flow Designer

Business Rules and Script Includes for logic implementation

Dashboard creation for visualisation

- **Operational Feasibility:** The system is designed to be simple, intuitive, and user-friendly. Employees can continue using the platform as part of their regular workflow without additional complexity.

Minimal learning curve for users already familiar with ServiceNow

Automated processes reduce manual effort

Clear dashboards help managers easily interpret data

Alerts and notifications ensure timely action

- **Economic Feasibility:** As an academic project developed on ServiceNow, the overall cost is minimal.

No need for additional hardware or infrastructure

Development is done using existing platform resources

Maintenance cost is low due to built-in automation

Can be scaled within organisations already using ServiceNow

3. PROJECT FEATURES

The Smart Personal Productivity & Burnout Detection System developed on ServiceNow includes the following key features:

- **User Roles & Access Control:**
 - **Employee/User Module** – Access to personal tasks, workload insights, and burnout score
 - **Manager/Admin Module** – Monitor team performance, workload distribution, and risk levels
 - Role-based access ensures secure and controlled data visibility
- **User Features:**
 - View and manage assigned tasks
 - Track number of active tasks and priority levels
 - Monitor personal productivity metrics
 - View real-time burnout score
 - Receive alerts when workload exceeds safe limits
- **Manager Features:**
 - Dashboard to monitor team workload and performance
 - Identify employees with high burnout risk
 - Analyse task distribution across team members
 - Take actions such as task reassignment or deadline adjustments
 - Access detailed reports and analytics
- **Burnout Detection System:**

Calculates a dynamic Burnout Score based on:

 - Active tasks
 - Task priority

Average completion time

Backlog trends

Categories users into:

Safe zone

Warning zone

Critical zone

- **Automation & Workflow Features:**

- Real-time alerts when burnout score crosses threshold
- Automated escalation workflows for continuous overload
- Notifications sent to both users and managers
- Task reassignment and workflow optimisation using Flow Designer

- **Dashboard & Reporting:**

- Interactive dashboards for employees and managers
- Visualisation of workload trends and productivity patterns
- Burnout risk indicators for quick decision-making
- Historical data analysis for performance improvement

- **Additional Features:**

- Complete activity and task history tracking
- Data-driven insights for better workload management
- Reduction of manual monitoring through automation
- Scalable and adaptable within organisational workflows
- Responsive interface for easy access across devices

4. TOOLS, TECHNOLOGIES AND SYSTEM REQUIREMENTS

Tools and Technologies Used:

Platform:

- ServiceNow (Core development platform)

Frontend:

- ServiceNow UI (Forms, Service Portal, Dashboards)
- HTML, CSS, JavaScript (for UI customization)

Backend:

- Business Rules (for server-side logic)
- Script Includes (modular backend scripting using JavaScript)
- Flow Designer (for automation workflows)

Database:

- ServiceNow Tables (Task, User, Custom Burnout Metrics Tables)

IDE / Development Environment:

- ServiceNow Studio
- VS Code (optional for scripting support)

Version Control:

- Git & GitHub (optional, for external code management)

Other Tools:

- Flow Designer (automation)
- Performance Analytics (for reports and dashboards)

Hardware Requirements:

- Processor: Intel Core i3 or above
- RAM: 8 GB or higher
- Storage: 250 GB or more
- Internet Connection: Required

Software Requirements:

- Operating System: Windows 10 / 11 / macOS

- Browser: Latest version of Chrome / Firefox / Edge
- ServiceNow Instance: Required

Functional Requirements

1. Users must be able to log in based on their roles (Employee / Manager / Admin)
2. Employees must be able to view and manage assigned tasks
3. System must track task-related data (active tasks, priority, completion time, backlog)
4. System must calculate and update the **Burnout Score** dynamically
5. Users must receive alerts when burnout thresholds are exceeded
6. Managers must be able to monitor team performance through dashboards
7. System must support task reassignment and escalation workflows
8. All user activity and task history must be recorded

Non-Functional Requirements

1. The system should be user-friendly with an intuitive interface
2. The system should provide real-time updates and notifications
3. Data should be stored securely with proper access control
4. System should be scalable within organisational environments
5. Response time for dashboards and alerts should be minimal (near real-time)
6. System should ensure high availability and reliability

5. WORK PLAN & METHODOLOGY

Methodology: The Smart Personal Productivity & Burnout Detection System is developed using a modular and iterative approach on the ServiceNow platform, ensuring flexibility, scalability, and easy integration of features.

The development is divided into the following key modules:

- **User Role & Access Management Module**

Implementation of role-based access (Employee, Manager, Admin) with secure login and controlled data visibility

- **Task Tracking & Management Module**

Integration with task records to monitor active tasks, priorities, deadlines, and completion status

- **Burnout Detection Module**

Design and implementation of logic to calculate the dynamic Burnout Score based on workload parameters

- **Automation & Alert System Module**

Use of Flow Designer to trigger real-time alerts, notifications, and escalation workflows when thresholds are exceeded

- **Dashboard & Analytics Module**

Creation of interactive dashboards for visualising workload distribution, productivity trends, and burnout risk levels

- **Database & History Management Module**

Storage and maintenance of task history, user activity, and burnout metrics for analysis and reporting

Project Timeline (January – May 2026)

Month	Phase / Task
January	Requirement Analysis, System Design, Database Structure, UI Wireframes
February	User Role & Task Management Module Development (Employee + Manager Setup)
March	Burnout Detection Module (Score Calculation Logic + Data Tracking)
April	Automation & Dashboard Module (Alerts, Flow Designer, Analytics Dashboard)
May	Testing, Debugging, Documentation, Final Deployment & Submission

Role & Responsibilities:

- **Aditi Garg:** Backend development, Database connectivity, and Logic implementation
- **Ankit Maheshwari:** Frontend design, UI/UX, and Page development
- **Jayesh Garg:** Logic implementation, Documentation and PPT

6. CONCLUSION

The Smart Personal Productivity & Burnout Detection System developed on the ServiceNow platform successfully demonstrates how intelligent monitoring and automation can enhance both productivity and employee well-being in modern work environments. Instead of limiting itself to traditional task tracking, the system introduces a proactive and data-driven approach to understanding workload patterns and identifying early signs of burnout.

By continuously analysing key parameters such as task volume, priority levels, completion time, and backlog trends, the system generates a dynamic Burnout Score that provides meaningful insights into an individual's workload pressure. This allows timely intervention through automated alerts, escalation workflows, and managerial oversight, ensuring that issues are addressed before they impact performance or health.

The integration of dashboards and analytics further strengthens decision-making by offering a clear view of team productivity and workload distribution. As a result, the system not only improves operational efficiency but also promotes a balanced and sustainable work culture.

In conclusion, this project highlights the importance of combining technology with human-centric design, proving that productivity and well-being can go hand in hand when supported by intelligent systems. It also lays a strong foundation for future enhancements such as AI-based predictions and deeper behavioural analysis, making it highly relevant for real-world organisational use.

